

Question

A compound with a garnet crystal structure contains 4 elements, with mass fractions of **A** 18.55%, **B** 14.52%, **C** 5.60%, **D** 61.33%, without considering non-stoichiometric compounds. The following descriptions exist:

1. The sum of the outermost electron numbers of **A, B, C, D** is 12.
2. There are 12 **A** atoms in a proper unit cell.
3. There are 16 **A** atoms in a proper unit cell.
4. There are 24 **C** atoms in a proper unit cell.
5. There are 96 **B** atoms in a proper unit cell.
6. According to Pauling's second rule, the charge number of the anion is equal to the sum of the surrounding ionic bond strengths.
7. There are transition elements among the four elements.
8. The sum of the outermost electron numbers of **A, D** is less than 8.
9. The coordination polyhedra of **A, B, C, D** in the unit cell do not have regular polyhedra.
10. The molecular weight of this compound is greater than 400.

Which of the following options describes a combination that does not contain incorrect descriptions?

A. All other options contain incorrect choices.

B. 2 5 7 9

C. 3 4 6 8

D. 1 4 6 9

E. 1 2 7 9

F. 1 4 6 8

G. 1 4 7 8 9

H. 1 4 5 9 10

I. 1 4 6 9 10

Answer

Correct Answer: **D**

Detailed Explanation

The general chemical formula of garnet is $X_3Y_2Z_3M_{12}$, where X, Y, Z are cations, and M is an anion. This general formula precisely contains 4 elements, and the ratio of their number of atoms (mole ratio) is fixed at 3 : 2 : 3 : 12.

CHECKPOINT

1 PTS

The ratio of the number of atoms of the 4 elements is 3 : 2 : 3 : 12

Determining the types of elements is a key step in solving the problem. Usually, we can assume that the element with the highest mass fraction is oxygen. However, if this assumption cannot lead to a reasonable chemical formula, other possibilities should be considered, such as fluorine, because the molar mass of fluorine is close to that of oxygen and it is also common in minerals. Verification shows that oxygen does not meet the conditions, so fluorine is most likely.

CHECKPOINT

1 PTS

The anion is most likely fluorine

Trying when $D = F$, the calculations show that A, B, C all have reasonable atomic weights. $D = F$, $A = Na$, $B = Al$, $C = Li$, it can be judged that description 7 is incorrect.

CHECKPOINT

2 PTS

A = Na, **B** = Al, **C** = Li, **D** = F, description 7 is incorrect

Relative molecular mass $M(\text{Na}_3\text{Al}_2\text{Li}_3\text{F}_{12}) = 371.75$, description 10 is incorrect.

CHECKPOINT

1 PTS

Molecular weight $M = 371.75 < 400$, description 10 is incorrect

The sum of the number of outermost electrons is $1 + 3 + 1 + 7 = 12$, description 1 is correct. The sum of the number of outermost electrons of Na, F is 8, description 8 is incorrect.

CHECKPOINT

1 PTS

The sum of the number of outermost electrons of **A**, **B**, **C**, **D** is 12, description 1 is correct, the sum of **A**, **D** is 8, description 8 is incorrect

In the garnet structure, $Z = 8$, so descriptions 2, 3, and 5 are incorrect, and description 4 is correct.

CHECKPOINT

1 PTS

One proper unit cell contains 24 **A**, 24 **C**, descriptions 2, 3, and 5 are incorrect, and description 4 is correct

In the garnet structure, all coordination polyhedra include $[\text{NaF}_8]$, $[\text{AlF}_6]$, $[\text{LiF}_4]$, $[\text{FNa}_2\text{AlLi}]$. According to Pauling's second rule, $\frac{1}{8} \times 2 + \frac{3}{6} + \frac{1}{4} = 1$, description 6 is correct.

CHECKPOINT

1 PTS

According to Pauling's second rule, $\frac{1}{8} \times 2 + \frac{3}{6} + \frac{1}{4} = 1$, description 6 is correct

In the unit cell, the symmetry of all atomic positions (16a, 24c, 24d, 96h) does not meet the requirements of a regular polyhedron (the highest 16a has only one $\bar{3}$ axis). Due to the different environments, there must be distortion, description 9 is correct.

CHECKPOINT

1.5 PTS

The symmetry of the atomic positions is lower than the symmetry of a regular polyhedron, there must be distortion, description 9 is correct

1 4 6 9 are correct, choose D